



Review

Optimizing exercise dosage for executive function in alzheimer's disease: A Bayesian dose-response meta-analysis of randomized trials

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HIGHLIGHTS

- This study modeled nonlinear dose-response effects of exercise on executive function in AD.
- The recommended dose is 670–1200 MET-min/week and aligns with WHO guidelines.
- Continuous aerobic training was effective at moderate loads for executive function.
- Integrated physical training required higher doses to improve executive function.
- Findings support precise, dose-based exercise prescriptions for executive dysfunction in AD.

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ABSTRACT

Objective: This study aimed to identify the optimal dose range of physical exercise for improving executive function in individuals with Alzheimer's disease (AD) and to compare the dose-dependent cognitive response profiles across different exercise modalities.

Methods: A systematic literature search was conducted across five databases-PubMed, Embase, Cochrane Library, SPORTDiscus, and Web of Science-up to June 20, 2025, with no language restrictions. All analyses were performed in R (version 4.4.2). A Bayesian framework was used to model continuous dose-response curves and to evaluate the relative effects of aerobic training, integrated physical training, and resistance exercise within a network structure.

Results: Exercise interventions demonstrated a nonlinear relationship with executive function, with the strongest effect observed at approximately 1000 MET-min/week. Among exercise types, continuous aerobic training showed significant improvement at 670 MET-min/week, indicating a relatively low-dose threshold for benefit. In contrast, integrated physical training programs required higher doses (around 1200 MET-min/week) to reach peak effect, suggesting greater intensity dependence. Resistance training did not yield consistent or statistically significant outcomes across the tested dose levels.

Conclusion: This study establishes a quantitative framework that integrates dose modeling with intervention structure to guide exercise-based cognitive strategies for AD. The findings emphasize the need to match exercise volume to modality demands and individual capacity, supporting the development of precision-oriented, non-pharmacological interventions. The identified optimal range (670–1200 MET-min/week) aligns with WHO recommendations for older adults, reinforcing both feasibility and translational potential.

1. Introduction

With global population ageing accelerating, the prevalence of neurodegenerative disorders is rising steadily ("2024 Alzheimer's disease

facts & figures," 2024a). Among them, Alzheimer's disease (AD) has emerged as a critical public health concern worldwide ("2023 Alzheimer's disease facts & figures," 2023). Characterised by progressive cognitive decline and functional impairment, AD imposes a growing

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burden on healthcare systems and families alike (Gaugler et al., 2022). Without effective intervention, the personal and societal costs of this disease are expected to escalate significantly in the coming decades ("2024 Alzheimer's disease facts & figures", 2024b).

Physical exercise has emerged as a safe and promising non-pharmacological intervention to support cognitive health in individuals with Alzheimer's disease (Paillard et al., 2024b; Sepúlveda-Lara et al., 2024). Mechanistically, exercise may enhance brain function by promoting the release of neurotrophic factors, improving synaptic plasticity, and reducing oxidative stress (Khemka et al., 2023; Wang et al., 2023; Yang et al., 2022). Accumulating evidence also suggests that regular physical activity is associated with a reduced risk of developing AD and may improve executive function in affected individuals (Ribarić, 2022).

Despite growing evidence for the cognitive benefits of exercise in AD, existing studies on executive function—a core cognitive domain—remain methodologically limited and fragmented. First, many studies combine executive function with other cognitive outcomes into global scores, obscuring its independent response to intervention and limiting mechanistic interpretation (Guo et al., 2023; Lv et al., 2023). Second, the diversity of exercise modalities used across studies has hindered systematic comparison, leaving open the question of which forms of exercise are most effective (Chen et al., 2024). Third, there is substantial variability in intervention dosage—such as frequency, intensity, and duration—without a consensus on optimal dose ranges, reducing comparability and limiting clinical translation (Ribas et al., 2023). Fourth, prior meta-analyses often include mixed samples of patients with mild cognitive impairment and AD, without clear diagnostic separation (Deng et al., 2024; Terao & Kodama, 2024). Given the distinct pathophysiological and neural substrates of executive dysfunction in AD, analyses targeting clearly diagnosed AD populations are urgently needed.

Exercise dose is typically defined as the product of frequency, intensity, and duration (i.e., frequency $\times$ intensity $\times$ time), and is commonly expressed in MET-minutes per week. It serves as a critical indicator for quantifying training load and evaluating whether the physiological demand matches the intended intervention goals. While prior studies have consistently demonstrated the cognitive benefits of exercise, especially in AD, few have systematically explored how much exercise is needed and whether different modalities interact with dosage to optimize outcomes. To address this gap, the present study focuses on individuals with clinically confirmed AD and targets executive function as the primary cognitive outcome. We introduce, for the first time, a unified Bayesian modeling framework that integrates both exercise dose and modality as core intervention dimensions (Swinton et al., 2024). At the aggregate level, we construct nonlinear dose–response curves to identify the minimum effective dose and optimal intervention range. At the comparative level, we examine the differential effectiveness of various exercise modalities across dose levels, revealing potential interactions between modality and training load. This dual-layer modeling approach overcomes limitations of previous meta-analyses by enhancing both precision and contextual interpretability. Ultimately, our findings aim to advance the field beyond the general conclusion that “exercise works” toward answering how it works, how much is needed, and in what form it is most effective, thus contributing to the development of individualized, data-informed exercise prescriptions for cognitive rehabilitation in AD.

2. Method

2.1. Protocol development and registration

A full review protocol was developed prior to the initiation of this study. The protocol was constructed in accordance with the PRISMA statement and its extension for network meta-analyses of healthcare interventions (Page et al., 2021), including the checklist and explanatory

documents (Hutton et al., 2015). The study was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration number: CRD420251081496), ensuring methodological transparency and procedural traceability.

2.2. Literature retrieval strategy

To comprehensively identify relevant studies, we conducted a systematic search across five major databases: PubMed, Embase, Cochrane Library, SPORTDiscus, and Web of Science. The search covered all publications up to June 20, 2025, with no language restrictions applied. Search terms and Boolean logic were adapted to fit the indexing characteristics of each database, balancing sensitivity and specificity. In addition, reference lists of systematic reviews and retrospective studies published in the past decade were manually screened to identify additional eligible records. Two reviewers independently conducted the initial screening and full-text assessments. Discrepancies were resolved through consultation with a third reviewer. The full search strategy, including keyword combinations, is available in Supplementary Table 1.

2.3. Eligibility criteria

Inclusion and exclusion criteria were defined based on the PICOS framework: Population, Intervention, Comparator, Outcomes, and Study design, to ensure systematic and reproducible selection.

2.3.1. Inclusion criteria were as follows

Population: We included studies involving participants with clinically diagnosed AD or confirmed through standardized cognitive assessments. A mean age of ≥ 60 years was required to ensure that the included populations reflected the typical age profile of AD onset and progression. This threshold enhances the clinical relevance of the findings by focusing on the demographic most likely to benefit from targeted cognitive interventions.

Intervention: Trials must have implemented structured physical exercise programs, including aerobic, resistance, coordination-based, or mind-body modalities, with clearly reported parameters (frequency, duration, and intervention period). Studies incorporating cognitive components alongside exercise were also eligible.

Comparator: Control groups were required to receive no structured physical activity intervention. Acceptable comparators included health education, routine behavioral maintenance, or light non-exercise activities (e.g., stretching or relaxation).

Outcomes: Studies had to report outcomes related to executive function, either through standardized scales or task-based measures, provided they specifically assessed executive processes.

Study design: Only published randomized controlled trials (RCTs) with clear randomization procedures and human participants were included.

2.3.2. Exclusion criteria were as follows

Studies that included non-AD populations or mixed samples (e.g., Parkinson's disease, cerebrovascular disease) without stratified reporting; Trials with a mean participant age below 60 years; Acute exercise studies involving a single bout without a structured training regimen; Control groups that included any exercise-based interventions; Non-randomized or quasi-randomized studies, including crossover designs; Studies lacking complete pre-post statistical data between groups, and for which data could not be obtained from authors; Studies with poor methodological quality (PEDro score < 4) (Fandim et al., 2022); Unpublished materials such as protocols, conference abstracts, dissertations, or other non-peer-reviewed sources.

2.4. Data collection process

Literature screening and data extraction were conducted

independently by two reviewers. Screening proceeded hierarchically from title review to abstract and full-text assessment, following the predefined eligibility criteria. For all included studies, core information was systematically extracted, including author and publication year, study design, sample size, participant characteristics, intervention details, and executive function assessment tools.

For quantitative data, means and standard deviations were the preferred extraction format. If not directly reported, authors were contacted to request original data. If unavailable, data were approximated from figures using WebPlotDigitizer software (Devron et al., 2017). When results were presented as intervals (e.g., confidence intervals, medians, or standard errors), established statistical formulas were applied to convert them into means and standard deviations (De Miguel-Rubio et al., 2023). When a single study reported multiple outcome measures for the same cognitive domain, we selected only one representative indicator to ensure statistical independence and avoid over-weighting that study in the meta-analysis. The selection process followed a predefined decision rule established prior to data extraction. Specifically, we prioritized measures that were (1) most frequently reported across all included studies, to enhance consistency and comparability; (2) supported by robust psychometric evidence from prior meta-analyses or clinical guidelines; and (3) most aligned with the dominant task paradigms in the field. In cases where multiple indicators were equally common and psychometrically sound, the final choice was based on the highest level of structural and conceptual comparability with other studies. This approach minimized selection bias and ensured methodological transparency and consistency in outcome extraction.

2.5. Methodological appraisal and bias control

The methodological quality of included studies was assessed using the Physiotherapy Evidence Database (PEDro) scale. Two reviewers independently scored each study, and discrepancies were resolved by consensus or through consultation with a third reviewer. The PEDro scale comprises 11 items, with the first item evaluating external validity and excluded from the total score. Scores range from 0 to 10, with classifications as follows: 4–5 indicating fair quality, 6–8 representing moderate to high quality, and 9 or above reflecting high methodological rigor. Studies scoring below 4 were considered methodologically inadequate and excluded from the quantitative synthesis (Amer-Cuenca et al., 2023; De Miguel-Rubio et al., 2023).

2.6. Statistical analysis

All statistical modeling and data analyses were performed using R software (version 4.4.2). A Bayesian nonlinear dose-response meta-analysis was constructed to quantify the relationship between exercise dose and improvements in executive function among individuals with AD (Mawdsley et al., 2016). Exercise dose was defined as the product of frequency, intensity, and duration, and treated as a continuous variable in the model (Etzioni & Kadane, 1995). Prior to modeling, the network structure was evaluated for basic assumptions required for network meta-analysis, including connectivity across interventions and the transitivity and consistency of the data. Initial visualizations of dose-effect trends were generated using effect estimates from the included studies (Wheeler et al., 2010; White et al., 2012). To accurately capture nonlinear response patterns, several functional forms were tested for model fitting, including Emax models, restricted cubic splines, quadratic functions, and non-monotonic curves (Pedder et al., 2019). Comparative performance was evaluated using the Deviance Information Criterion (DIC), model complexity, between-study heterogeneity, and residual fit (Evans, 2019). The quadratic function outperformed all other models across these dimensions and was selected for final modeling.

Bayesian modeling was conducted using Markov chain Monte Carlo (MCMC) simulations to estimate posterior distributions, from which

effect estimates and 95 % credible intervals (CrIs) were derived. An effect was considered statistically significant when the CrI did not cross zero (Etzioni & Kadane, 1995). For consistency in outcome direction, studies using reverse scoring (e.g., lower scores indicating better function) were adjusted accordingly. Model implementation was conducted using the “MBNMAdose” package, and visualizations were produced via the “ggplot2” package, yielding nonlinear dose-response curves. Additionally, the certainty of evidence for primary outcomes was assessed using the GRADE (Grading of Recommendations, Assessment, Development and Evaluation) framework. This evaluation considered the risk of bias, consistency, directness, precision, and potential publication bias to determine the overall credibility and applicability of the findings (Kavanagh, 2009).

In contrast to traditional frequentist meta-analytic approaches, Bayesian methods offer several advantages that are particularly well-suited for complex dose-response modeling (Hamza et al., 2021). First, the Bayesian framework enables the incorporation of prior knowledge and provides full posterior distributions of model parameters, which facilitates probabilistic interpretation and quantification of uncertainty. Second, Bayesian models are inherently flexible, allowing for the estimation of non-linear dose-response relationships and hierarchical structures without relying on asymptotic approximations. Third, Bayesian methods are especially advantageous in scenarios with sparse or heterogeneous data-common features in clinical trials involving older adults with AD, because they yield more stable inferences through shrinkage and prior regularization. By adopting a Bayesian dose-response network meta-analysis, the present study is able to simultaneously compare multiple exercise modalities across various dose levels and generate clinically interpretable estimates with credible intervals that reflect true uncertainty.

3. Results

3.1. Literature selection

A comprehensive search across multiple databases was conducted, followed by a structured multi-stage screening process. Ultimately, 17 studies were included for both qualitative and quantitative synthesis (Arcoverde et al., 2014; Budak et al., 2023; David et al., 2025; Enette et al., 2020; Fonte et al., 2019; Henskens et al., 2018; Hoffmann et al., 2016; Holthoff et al., 2015; Lautenschlager et al., 2008; Morris et al., 2017; Öhman et al., 2016; Papatsimpas et al., 2023; Silva et al., 2019; Stein et al., 2023; Toots et al., 2017; Yáñez et al., 2011; Yu et al., 2021). The full selection process is illustrated in the PRISMA flow diagram (Fig. 1).

3.2. Study characteristics

The 17 included studies were published between 2008 and 2025, comprising a total of 1377 older adults with clinically confirmed Alzheimer's disease. Among them, 716 participants were assigned to exercise intervention groups, and 678 to control groups. All participants were aged over 60 years, and diagnoses were based either on established clinical criteria or validated Alzheimer's-specific assessment tools. Intervention durations varied from 2 to 48 weeks, with the longest intervention appearing in only one study. Most studies employed a 12-week protocol, with an average intervention duration of approximately 16 weeks. Exercise frequency ranged from two to five sessions per week, most commonly between two and three sessions weekly. Control groups typically received standard non-exercise care, such as health education or psychosocial support. All exercise programs were classified by experts in exercise science, encompassing a variety of well-defined training modalities. The present review focused on three core domains of executive function: cognitive flexibility, working memory, and inhibitory control. These domains are known to play a critical role in maintaining autonomy and daily functioning in individuals with

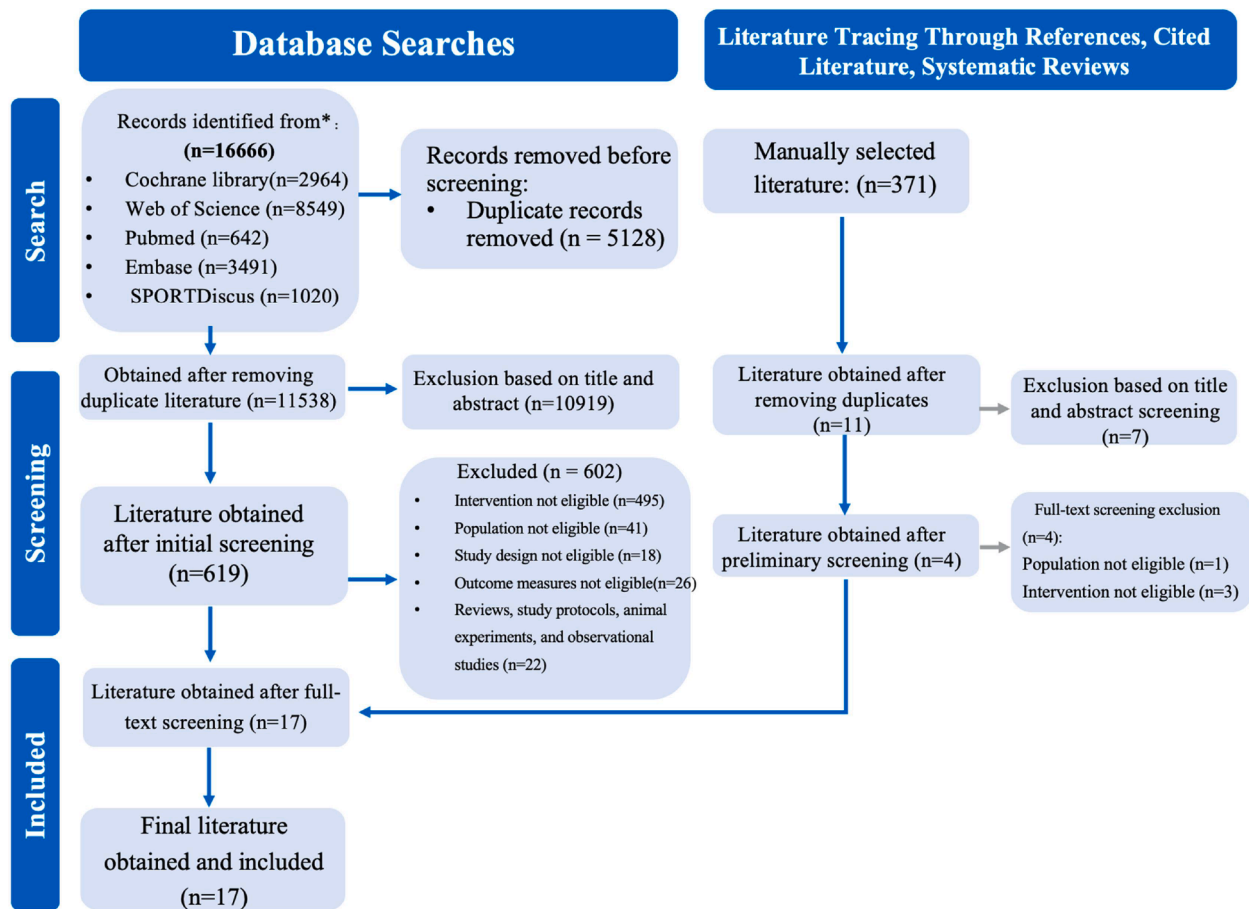


Fig. 1. PRISMA flow diagram of study selection.

Alzheimer's disease (Cammisuli et al., 2024; Pisano et al., 2020; Wright et al., 2023). Several studies reported minor adverse effects during the early phases of the intervention, including delayed-onset muscle soreness, mild joint discomfort, and occasional falls. However, these were transient, did not interrupt intervention delivery, and were likely attributable to participants' low baseline physical activity levels and age-related functional decline. Key details of the included studies and their respective interventions are summarized in Table 1.

To enhance clarity, the exercise interventions in the included studies were categorized into three modalities based on their original descriptions: continuous aerobic training, integrated physical training, and resistance training. Continuous aerobic training referred to single-form, uninterrupted cardiovascular activities such as walking, cycling, or treadmill training conducted at a steady moderate intensity. Integrated physical training included multicomponent interventions that combined at least two distinct types of physical activity (e.g., aerobic + resistance, or aerobic + balance), often delivered in structured circuits or alternating formats. Resistance training interventions were strength-focused protocols involving bodyweight exercises, elastic bands, free weights, or machine-based training, aiming to improve muscular strength or endurance.

3.3. Risk of bias assessment

All included studies underwent methodological quality assessment. The overall study quality was moderate to high, with a mean score of 6.3 out of a possible maximum. Thirteen studies achieved scores of 6 or above, indicating generally sound design. Ten studies employed blinded outcome assessment to minimise detection bias. Due to the nature of exercise interventions, blinding of participants and interventionists was

not feasible—an inherent limitation in non-pharmacological trials. However, as the outcome measures were primarily performance-based tasks, the absence of participant blinding was considered to exert minimal influence on the results. Only one study reported an attrition rate exceeding 15 %. This study addressed missing data using intention-to-treat analysis, thereby mitigating the risk of attrition bias. Collectively, the studies demonstrated satisfactory protocol adherence, high participant compliance, and effective risk management, reflecting the feasibility and acceptability of exercise interventions in this clinical population.

3.4. Posterior distribution analysis of the effects of different exercise doses and modalities

To assess the stability of the Bayesian model and the performance of the credible intervals, posterior distributions were generated for different exercise doses and intervention modalities relative to the placebo, as shown in Fig. 2. Fig. 2A illustrates that overall exercise interventions demonstrated a positive treatment effect across all dose levels (150–1500 MET-min/week). The posterior density distributions for the 500 and 750 MET-min/week doses were the most concentrated, with 95 % credible intervals (CrIs) showing good convergence, supporting these doses as some of the optimal ranges for effect. Fig. 2B further illustrates the posterior effect distributions for different exercise modalities at their respective doses. Continuous aerobic training and integrated physical training exhibited clear positive effects at moderate to high doses (e.g., 750–1250 MET-min/week), with sharp density curves skewed toward the positive region, indicating strong intervention potential for executive function. In contrast, resistance training showed a flatter distribution, with its CrI crossing zero, suggesting that its effects

Table 1
Characteristics of the included studies.

Source	Study design	Diagnostic standards	Sample size	Gender (M,F)	Age, Mean±SD	Intervention type (E)	Weeks/Frequency per week/Duration-min	Codes/METs-min	Outcome
Nicola T. Lautenschlager, 2008	R-T-S	ICD-10	85/85	43,42/41,44	68.6 ± 8.7/68.7 ± 8.5	ART	12/3/50	0206,260/5.0	CDR, DST, VFT
Lidia Yaguez, 2011	R-T	ICD-10	15/12	6,9/5,7	70.5 ± 8/75.7 ± 6.90	IPT	6/NA/120	2240,060/5.5	CNTAB
Cynthia Arcoverde, 2014	R-T-S	CDR	10/10	5,5/4,6	78.5/79	JT	12/2/30	1202,860/8.5	VFT, TMT, ST, DST
Vjera A. Holthoff, 2015	R-T-S	CDR	15/15	7,8/8,7	72.40±4.34/70.67±5.41	RT	12/3/30	0205,160/3.0	SWFT
Kristine Hoffmann, 2016	R-T-D	MMSE	107/93	57,36/56,51	69.8 ± 7.4/71.3 ± 7.3	CYT	16/3/60	0101,760/4.5	DST, NI
Hannareeta Ohman, 2016	R-T	DD	70/70	45,25/44,26	78.3 ± 5.1/78.1 ± 5.3	IPT	48/2/120	0206,260/5.0	CDT, VFT
Jill K. Morris, 2017	R-T-S	CDR	39/37	18,21/21,16	74.4 ± 6.7/71.4 ± 8.4]	IPT	26/NA/150	0206,260/5.0	DST
Annika Toots, 2017	R-T-S	DD	93/93	23,70/22,71	84.4 ± 6.2/85.9 ± 7.8	IPT	16/2/45	0206,260/5.0	VFT
Marinda Henskens, 2018	R-T-S	DD	22/22	5,17/5,17	85.14±4.64/84.73±4.55	IPT	24/3/45	0206,260/5.0	DST, FAB
Cristina Fonte, 2019	R-T-S	DD	19/20	7,12/6,14	79±7/79±9	IPT	26/3/90	0206,260/5.0	FAB, DST, TMT
Felipe de Oliveira Silva, 2019	R-T-S	CDR	14/13	3,11/8,5	77.54±8.05/81.22±8.88	IPT	12/2/60	0206,260/5.0	CDR, CDT, VFT, ST
Lievyne Enette, 2020	R-T	DD	17/21	6,11/6,11	79	CYT	9/2/30	0101,760/4.5	DST
Fang Yu, 2021	R-T	CDR	64/32	36,28/17,15	77.4 ± 6.6/77.5 ± 7.1	CYT	2/3/50	0101,760/4.5	DST, TMT, CDT
Vasileios Papatsimpas, 2023	R-Th-D	DD	57/57/57	21,36/17,40/7,50	76.82±5.73/76.07±5.7/78.75±7.06	ART/RT	12/3–5/30–45	0206,260/5.0, 0205,160/3.0	TMT, DST
Miray Budak, 2023	R-Th	CDR	9/9	5,4/1,7	73.60±8.62/74.90±8.11	IPT	2/5/50	0101,760/4.5	CDT, VFT, DST, ST
Angelica Miki Stein, 2023	R-T	CDR	18/16	7,9/8,10	75.06±6.36/75.27±6.09	JT	12/3/25–40	1202,860/8.5	CDT, VFT, FAB
Shari David, 2025	R-T	CDR	22/19	13,9/12,7	72.1 ± 5.8/68.0 ± 8.2	ART	26/2/45	0206,260/5.0	MoCA

Notes: E, Experimental group; C, Control group; NA, Not applicable; ICD-10, international Classification of diseases; CDR, Clinical Dementia Rating; MMSE, Mini Mental State Examination; DD, Dementia diagnosis; R, Randomized controlled trials; T, Two-armed; Th, Three-armed; S, Single-blind; D, Double-blind; ART, Aerobic and Resistance; ECCT, Exercise combined with cognitive training; JT, Jogging; RT, Resistance training; IPT, Integrated physical training; CYT, Cycling; DST, Digit Span test; VFT, Verbal fluency test; CNTAB, Cambridge Neuropsychological Test Automated Battery; TMT, Trail Making Test; ST, Stroop Test; SWFT, Semantic word fluency test; NI, Neuropsychiatric Inventory; CDT, Clock Drawing Test; FAB, Frontal Assessment Battery.

were unstable or potentially not statistically significant. The solid and dashed lines represent direct and indirect evidence distributions, respectively, confirming consistency in effect estimates across different paths in the network model.

3.5. Dose-Response relationship of overall exercise dose on executive function in Alzheimer's disease

The Bayesian model results revealed a nonlinear dose-response relationship between overall exercise intervention and improvements in executive function in Alzheimer's disease patients. Fig. 3A displays the network comparison structure of overall exercise at various dose levels against the placebo, while Fig. 4A presents the corresponding dose-effect curve. Intervention effects were significant starting at approximately 170 MET-min/week, and continued to increase with higher doses, with the most pronounced improvements observed between 170 and 670 MET-min/week. The effect peaked at 1000 MET-min/week (SMD = 0.248, 95 % CrI: 0.039–0.517), which represented the highest dose with statistically significant improvements. Beyond this dose, the CrI crossed zero, indicating that the effect plateaued and ceased to be statistically significant. Specific estimates for each dose level are provided in Supplementary Table 6. Further ranking probability analysis, shown in Fig. 5A, indicated that among all overall exercise doses, 1000 MET-min/week had the highest Rank 1 probability, suggesting that this dose was most likely to rank first in the network

comparison, offering the strongest overall advantage. These findings align with the dose-response modeling conclusions, further supporting 1000 MET-min/week as the optimal dose point.

3.6. Comparison of dose-response relationships for different exercise modalities

Based on Bayesian network modeling, different exercise modalities exhibited nonlinear dose-response relationships in improving executive function in individuals with Alzheimer's disease. As shown in Fig. 4B, continuous aerobic training demonstrated the most significant and statistically meaningful intervention effect at 670 MET-min/week (SMD = 0.251, 95 % CrI: 0.003–0.555), which was identified as the optimal dose for this modality. Higher doses, such as 330 or 420 MET-min/week, showed higher mean effects, but their lower bounds of the credible intervals fell below zero, making them statistically insignificant. Integrated physical training reached its peak effect at 1200 MET-min/week (SMD = 0.285, 95 % CrI: 0.026–0.597), with steady improvement across the entire moderate-to-high dose range (500–1300 MET-min/week), and the credible intervals rarely crossed zero, suggesting strong intervention potential and dose consistency. Resistance training showed the highest effect at 190 MET-min/week (SMD = 0.276), but the 95 % CrI crossed zero at all dose levels, indicating an unstable effect and substantial uncertainty regarding its efficacy.

In summary, continuous aerobic training showed significant

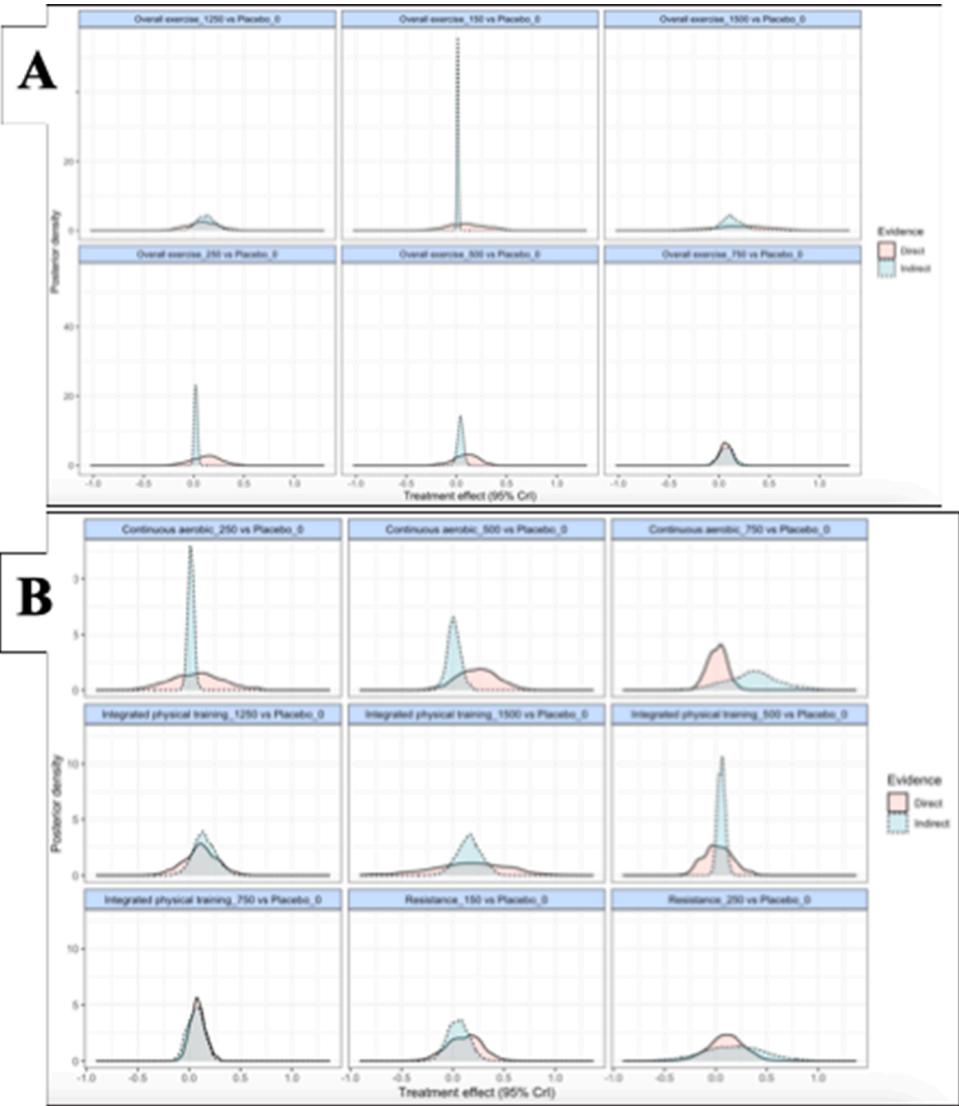


Fig. 2. Posterior dose–response distributions of exercise effects on executive function: (A) overall exercise dose compared to placebo; (B) modality-specific exercise effects across corresponding doses compared to placebo.

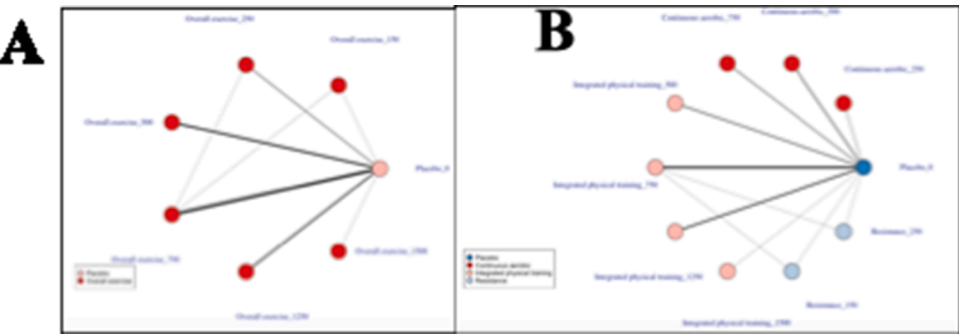


Fig. 3. Network plots of comparisons among exercise modalities and doses versus placebo.

improvements at moderate doses (670 MET-min/week), integrated physical training reached its optimal effect at higher doses (1200 MET-min/week), while resistance training lacked clear evidence of efficacy within the current dose range. Detailed data are provided in Supplementary Table 7, and dose-response curves are presented in Fig. 4B As shown in Fig. 5B, in comparisons of specific exercise modalities and dose combinations, integrated physical training at 1200 MET-min/week had

the highest Rank 1 probability, followed by continuous aerobic training at 670 MET-min/week, suggesting these two combinations are positioned optimally in the network comparison. This result further underscores the critical role of modality and dose matching in intervention effectiveness.

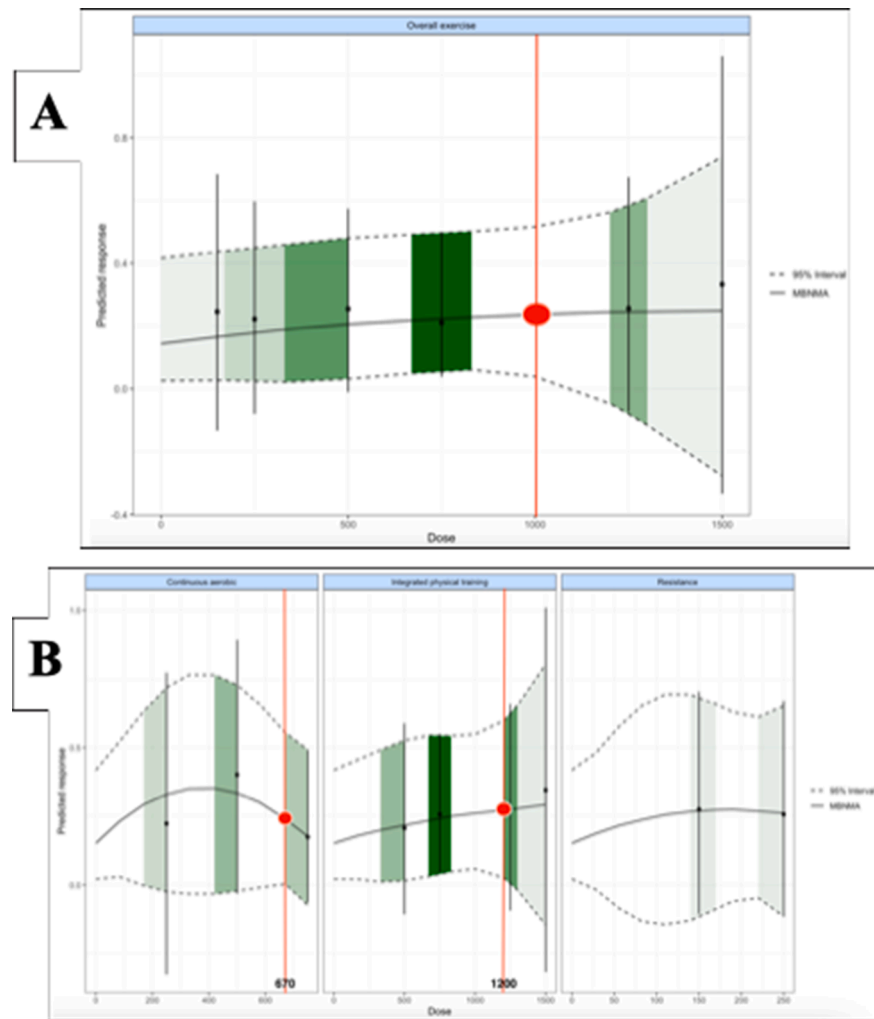


Fig. 4. Estimated dose-response relationships for overall (A) and modality-specific (B) Exercise Interventions compared to placebo, based on model-based network meta-analysis.

3.7. Sensitivity analysis: dose estimation stability and evidence quality assessment

To assess the robustness of the Bayesian modeling results and the reliability of the conclusions, Supplementary Table 8 presents a systematic evaluation of the evidence quality for the improvement of executive function in Alzheimer's disease patients through exercise interventions, using the GRADE system. The results indicated that while most studies had issues with allocation concealment and blinding (which led to downgrading for risk of bias), the findings were consistent in terms of directionality, and the study populations and outcome measures were well-matched. No significant indirectness or publication bias was observed. Considering some studies had small sample sizes and wide 95 % credible intervals, the overall evidence was rated as "moderate quality."

To further test the robustness of the dose-response relationship, Supplementary Figures 1–4 show the fitting trends of overall exercise and the three primary exercise modalities (continuous aerobic, integrated physical training, and resistance training) at various dose levels, compared with actual observed values. The results indicated that all fitting curves adequately reproduced the dose-effect trajectory, and the model consistently captured the trends of effect changes and peak positions, further supporting the stability and explanatory power of the model.

Finally, Supplementary Figures 5–6 used funnel plots and Sunset

plots to assess publication bias and statistical power. The Egger test found no significant small sample effects ($z = -0.326$, $p = 0.744$), indicating the robustness of the conclusions. In summary, the modeling results demonstrated strong stability across multiple dimensions of sensitivity and validation analysis, providing a reliable basis for future practice.

4. Discussion

4.1. Key findings: exercise dose and cognitive benefit threshold in Alzheimer's disease

This study used a Bayesian approach to construct a continuous dose-response model, systematically evaluating the effects of exercise intervention on executive function at varying intensity levels in AD patients. The analysis revealed that the positive effects of total exercise volume on cognition do not continue to increase indefinitely but instead peak at around 1000 MET-min/week, after which the benefits plateau or even show diminishing returns. This threshold is supported not only by effect size estimates but also by posterior distributions and ranking probabilities, which indicate high concentration and a leading rank for this dose. This finding significantly complements previous research on exercise interventions that lacked fine-grained dose control. Most prior protocols derived intervention volumes from general healthy populations, overlooking the unique neurophysiological and functional reserve

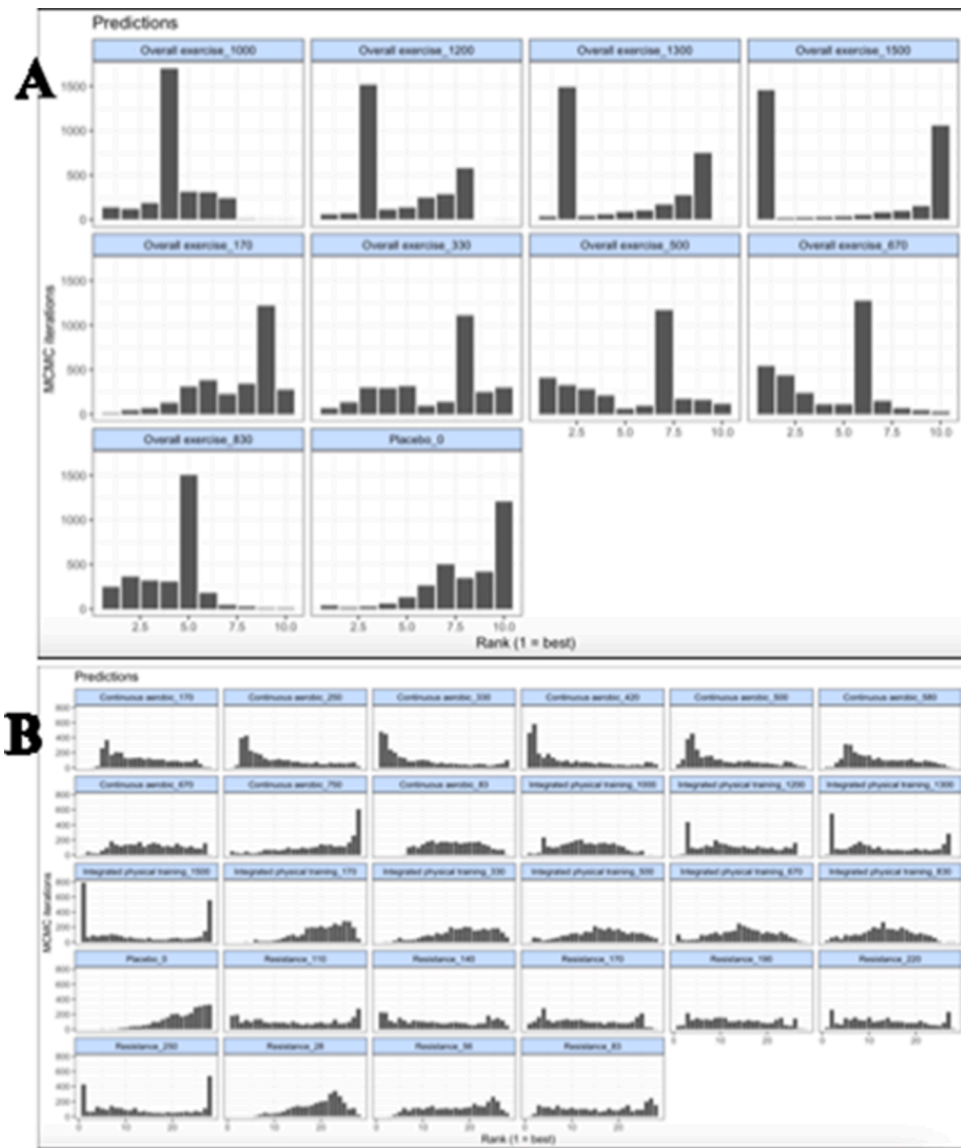


Fig. 5. Rank probabilities of exercise interventions compared with placebo: (A) overall exercise across different dose levels; (B) modality-specific interventions across corresponding dose combinations.

considerations of AD patients. Through precise modeling and continuous dose localization, this study provides high-resolution data support, clearly identifying a defined exercise threshold and optimal response range for the AD population, laying the groundwork for differentiated exercise prescriptions. Although this study focuses specifically on individuals with AD, it is important to clarify that AD is only one of the most common forms of dementia and does not represent all dementia conditions. Other types, such as vascular dementia and Lewy body dementia, may share certain features with AD, particularly in terms of executive dysfunction and neurodegenerative changes, but differ in their underlying pathologies and trajectories of decline. Therefore, the dose response relationship established in this study should not be directly generalized to other dementia populations. Future research should investigate whether different dementia types present distinct dose thresholds to inform more targeted and individualized exercise intervention strategies.

From a neuroregulatory perspective, moderate exercise may improve task regulation and cognitive flexibility by activating the prefrontal-hippocampal pathways, enhancing neurotrophic factor levels, and optimizing brain perfusion and energy metabolism (Camara-Calmaestra et al., 2022; Delgado-Peraza et al., 2023; Li et al., 2023; Xu et al., 2023). However, when the

intervention dose exceeds the physiological adaptive range, inflammatory responses, cortical inhibition dysfunction, or neuronal resource misallocation may occur, limiting the persistence of positive effects (Deng et al., 2025; Persson et al., 2013). Unlike traditional "one-size-fits-all" recommendations, this study emphasizes the need to define intervention load thresholds based on the neurofunctional state and capacity limits of individuals with Alzheimer's disease (Afanador-Restrepo et al., 2024; Oliveira et al., 2024). Specifically, when developing a periodic exercise plan, approximately 1000 MET-min/week should be considered a key reference interval. By using phased progression and dynamic adjustment strategies, cognitive benefits can be maximized while minimizing the risk of neurofatigue and overload, thereby advancing intervention practices toward higher levels of precision and individualization.

4.2. Exploring effect thresholds and structure-load adaptation of different exercise modalities

The study shows that different types of training interventions exhibit systematic differences in dose-response patterns in improving executive function among AD patients, suggesting inherent divergence in

activation pathways and load adaptation mechanisms (Aczel et al., 2022; Wang et al., 2025b). Continuous aerobic training demonstrated significant cognitive improvements at a dose of 670 MET-min/week, indicating that its neurophysiological mechanisms have a lower activation threshold. Previous studies have shown that aerobic activity can enhance blood flow and metabolic states in the prefrontal cortex, accompanied by increased release of neurotrophic factors, which contribute to the overall efficiency of the cognitive control system (Norevik et al., 2024; Ribarić, 2022). These processes are typically triggered at moderate load levels, making this modality particularly suitable for older populations and those with functional decline (Li et al., 2024; Paillard et al., 2024a).

In contrast, integrated physical training showed the most significant effects when the dose was raised to approximately 1200 MET-min/week. The improvement pathway likely depends on more complex, coordinated neural regulation. This training type typically involves multidimensional body-cognition tasks, such as response inhibition, spatial coordination, and task-switching, demanding greater attention resources and multiregional brain network activation. Therefore, only when intensity and frequency reach a certain threshold can these functional networks be effectively activated. The greater the training complexity, the more reliant it is on load accumulation, suggesting that this intervention requires a more optimized combination of intensity and structure (Chang et al., 2022; Wang et al., 2025a).

Resistance training, however, did not show stable intervention advantages in this study, possibly due to its primarily indirect effects on the brain through muscle metabolism regulation or peripheral hormonal signaling, this mechanism may have limited responsiveness in neurodegenerative conditions. Although this study identified nonlinear dose-response relationships across exercise modalities, these findings do not imply that exercise modality itself is irrelevant. In fact, the structural characteristics inherent to each modality, such as cognitive demand, motor complexity, and coordination requirements, may interact with dose intensity to influence intervention efficacy through distinct neurophysiological mechanisms. Thus, it is not modality or dose alone, but the combination of both that likely determines the observed outcomes. It is important to note, however, that each modality category may include diverse implementation forms. For example, walking, cycling, and treadmill training were all classified under “Continuous aerobic.” Our analysis was constrained by the level of detail reported in the original studies, and therefore, we were unable to further examine within-modality heterogeneity or modality-specific dose adaptations. This limitation reflects a broader challenge of meta-analytic approaches, which often lack the granularity required to disentangle the complex interactions among intervention structure, training load, and underlying mechanisms. Future research using individual participant data (IPD) and detailed intervention reporting is needed to clarify these interactions and inform the development of more precise and mechanism-based exercise prescriptions (Gavelin et al., 2021; Jia et al., 2025).

4.3. Clinical implications and methodological innovation

This study developed a dose response model for exercise interventions in individuals with AD, demonstrating both methodological innovation and practical significance. Unlike previous studies that relied on broad categorical comparisons or uniform dosage assumptions, this study introduced a continuous dose effect function to characterize the nonlinear relationship between total exercise volume and improvements in executive function. The analysis identified the most robust cognitive benefits within the range of 670 to 1200 MET-minutes per week. Notably, this dose range closely aligns with the physical activity guidelines issued by the World Health Organization, which recommend that older adults, including those with chronic conditions or cognitive decline, engage in 150 to 300 min of moderate-intensity aerobic activity per week (Bull et al., 2020). This duration corresponds to approximately 600 to 1200 MET-minutes per week, depending on the exercise modality

and intensity. Therefore, the optimal threshold identified in this study is both practically attainable for individuals with mild to moderate AD and consistent with international public health recommendations. When implemented through structured, low-impact activities such as walking, tai chi, or supervised aerobic programs, this volume provides a realistic and evidence-based foundation for individualized exercise prescription in cognitive rehabilitation.

More importantly, this study systematically reveals the dynamic adaptation between intervention modality and load requirements. Continuous aerobic training produces beneficial effects at moderate load levels, making it suitable as a foundational exercise for individuals with limited physical capacity. In contrast, more complex integrated physical training requires higher doses to effectively impact areas such as multitasking and attention control. This differentiated response framework suggests that modality selection should not be made in isolation but must be assessed in conjunction with its neurophysiological stimulation characteristics and the adaptive load.

Overall, this research provides actionable, generalizable empirical support in three key areas: dose modeling methodology, identification of optimal dosage ranges, and intervention strategy stratification. Compared to previous studies that categorized interventions based on the presence or intensity of exercise, this research closely aligns with the practical needs of intervention implementation and holds potential for direct translation into clinical practice. The findings are significant for optimizing exercise intervention pathways in neurodegenerative diseases and offer valuable insights for related research in methodology and study design.

4.4. Methodological features and study limitations

This study demonstrates several advantages in both intervention modeling strategy and practical applicability. First, a continuous Bayesian dose-response model was used to examine the relationship between total exercise dose and cognitive performance, overcoming the limitations of traditional grouped analysis in dose determination. This approach enhanced the resolution and interpretative depth of intervention dose evaluation. Second, the effective dose range for improving executive function was clearly defined, providing a quantifiable reference for intervention and laying the empirical foundation for developing more targeted exercise programs. Additionally, the study accounted for the interaction between exercise modality and dose during modeling, revealing the heterogeneity in load requirements and intervention effects across different exercise types, thus providing a basis for modality selection in practical applications.

Despite these strengths, several limitations should be noted. First, there were inconsistencies in the definition, units, and calculation methods for exercise dose across the included studies. While standardization was performed for modeling, potential bias in dose estimation accuracy remains. Second, the study focused exclusively on executive function as a cognitive outcome, without addressing other cognitive domains in Alzheimer's disease, such as memory, language, or spatial abilities, which limits the generalizability of the findings. Third, the categorization of exercise modalities was primarily based on the original study descriptions, and some training programs exhibited overlap, potentially affecting the comparability of different modalities.

Another limitation of this study lies in the use of a mean age threshold (≥ 60 years) to define the older adult population. Although this criterion was intended to focus on the demographic most affected by AD and ensure the clinical relevance of the intervention, using study-level mean age may obscure substantial variability in actual age composition across samples. In particular, studies with large standard deviations may have included middle-aged participants despite reporting a mean above the threshold, thereby introducing within-study heterogeneity. This uncontrolled age variation may affect the estimation stability of dose-response relationships, especially for outcomes such as executive function that are sensitive to age-related cognitive decline.

Furthermore, the structure of aggregate data used in this analysis prevents us from examining whether different age subgroups (e.g., younger-old vs. older-old) respond differently to exercise interventions. Future trials should report more complete age-related information, including minimum and maximum values and distributional characteristics, to enable more precise moderator analyses in evidence synthesis.

Overall, this study provides a systematic exploration of intervention efficacy evaluation, dose-modality integration, and methodological paths, offering a reference for future efforts to optimize personalized cognitive intervention strategies. Future research could further improve the consistency of exercise dose measurement standards, expand cognitive outcome assessment dimensions, and enhance the standardization of intervention content to strengthen model interpretability and clinical applicability.

5. Conclusion

This study developed a comprehensive analytical framework combining continuous dose modeling and exercise modality evaluation to explore the optimal intervention strategy for enhancing executive function in Alzheimer's disease patients. The analysis identified a key exercise dose range of approximately 670 to 1200 MET-min/week for significant cognitive improvement and revealed distinct differences in load requirements and intervention outcomes across different exercise modalities. Aerobic training produced positive effects at moderate load levels, while integrated physical training, though requiring higher exercise volumes, showed greater intervention potential in individuals with better baseline functional capacity. The analytical approach proposed in this study offers a feasible pathway for developing precise, stratified cognitive training programs and lays the foundation for transitioning exercise interventions from broad recommendations to data-driven, structured, and individualized strategies. Future research should expand cognitive outcome measures and validate the model in real-world settings to enhance the feasibility and broader applicability of intervention strategies in clinical practice.

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Sihong Sui: Writing – original draft, Formal analysis, Data curation.
Mingyi Wang: Validation, Supervision, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare no conflict of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.archger.2025.106001](https://doi.org/10.1016/j.archger.2025.106001).

Data availability

All data analyzed in this study were extracted from previously published studies included in the meta-analysis. No new data were generated.

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